

Xode Staking and Node Architecture

By: Engr. Harold Glenn Minerva, ECE

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Abstract

Xode Blockchain is the Philippines' first Layer-1 blockchain platform, operating as a Polkadot parachain (**Para ID 3417**). Built on Substrate using the Cumulus framework, Xode combines smart contract support, multi-asset capabilities, and decentralized finance, with a native Proof-of-Stake consensus mechanism secured by XON tokens.

This whitepaper presents the technical architecture and economic model of Xode's staking system, covering both Author (Collator) Nodes and Xaver Nodes - the two distinct node types that together underpin the network's block production and reward infrastructure. Author Nodes are operated by independent node operators with bonds, author blocks, and earn rewards through the annual 4% XON staking release as well as 80% of the gas fees from blocks they produce.

Xaver Nodes are exclusively maintained and operated by the Xode Foundation as reserve node infrastructure, accepting XAV token delegations and distributing 100% of the 80% share of mined gas fees proportionally to delegators based on their fixed XAV tier share. The document also formalizes the mathematical models governing reward computation, delegation mechanics, XON-to-XAV conversion, slashing conditions, and the overall tokenomics of the XON staking ecosystem.

1. Introduction

1.1 Background

Decentralized networks require economic incentives to sustain participation in block production and validation. Xode Blockchain implements a Nominated Proof-of-Stake (NPoS) inspired model, adapted for a Polkadot parachain context where collators - rather than validators - are responsible for producing parachain blocks.

The native token of Xode is XON, with a total supply of approximately **122,493,748.83 XON**. A dedicated staking allocation of **60%** of the total supply carries an annual reward rate of 4%, creating sustainable, long-term incentives for network participants.

1.2 Author Node

An Author Node is an independent collator node operated by a third-party node operator on the Xode network. Author Nodes are responsible for collecting pending transactions, assembling parachain blocks, and submitting Proofs of Validity (PoVs) to the Polkadot relay chain for finalization.

To participate, an operator must bond 100,000 XON (Xode Foundation Bond Provision Program), which is locked on-chain as a security deposit for the duration of their active candidacy. In return, Author Nodes earn two distinct streams of reward: a proportional share of the annual 4% XON staking release, and 80% of all gas fees collected from every block they produce. The remaining 20% of gas fees is remitted to the on-chain treasury.

Author Nodes can accept delegations from XON token holders. The node operator may set a commission rate, which is deducted from the node's total rewards before distribution. The remaining rewards are then distributed to delegators proportionally, based on each delegator's share of XON staked relative to the total XON delegated to that node. Formally, the reward for a delegator is calculated as:

$$\text{Delegator Reward} = (\text{Delegator XON Staked} / \text{Total XON Delegated to Node}) \times (\text{Node Reward} \times (1 - \text{Commission Rate}))$$

This model ensures that larger delegations receive proportionally greater rewards, while the operator's commission provides a configurable incentive for node operators to attract and retain delegators. Collator selection each session is determined by **pallet-xode-staking** based on total effective stake - the sum of the operator's own bond and all delegated XON - ensuring that block production remains competitive and permissionless.

1.3 Xavier Nodes

Xaver Nodes are a specialized class of reserve nodes within the Xode network. Unlike standard collator nodes that require manual operator intervention, Xaver Nodes are designed with the following distinguishing properties:

- Automatic staking - Xaver Nodes self-stake without requiring manual bonding transactions from operators.
- Delegation acceptance - token holders can delegate XON to a Xaver Node and share in its staking rewards. Delegation is supported through two participation methods:
 - **Xaver Mining App:** Users who prefer a fully digital participation model can access delegation through the Xaver Mining App. This software application allows token holders to register, select an active Xaver Node, specify their delegation amount, and begin earning staking rewards - all without any physical hardware. The app provides a user-friendly interface for monitoring delegation status, accrued rewards, and node performance in real time.
 - **Xaver Lite Node:** Users who prefer a physical, dedicated device can purchase a Xaver Lite Node Appliance - a compact hardware unit purpose-built for Xode

network participation. The Xaver Lite Node runs a lightweight node client locally, automatically connecting to the Xode network and managing delegation on behalf of the owner. This method is suited for users who want a dedicated, always-on device that operates independently of a general-purpose computer or mobile application, providing greater uptime reliability and a tangible representation of their network participation.

- Proportional reward distribution - rewards are sourced from Xode gas fee mining. Of the total gas fees collected per block, 80% goes to the node that authors the block and 20% goes to the treasury. 100% of the Xode gas fees earned by an Xaver Node are distributed to delegates proportionally based on their locked XAV tokens, with no portion retained by the node operator.

2. Architecture

2.1 Parachain Context

As a Polkadot parachain, Xode relies on the relay chain for shared security and finality. Collator nodes on Xode are responsible for collecting transactions, building parachain blocks, and submitting Proofs of Validity (PoVs) to the relay chain. The staking system governs which nodes are eligible to serve as active collators in any given era.

2.2 Staking Pallet Architecture

The Xode staking system is implemented as a single, purpose-built Substrate FRAME pallet: **pallet-xode-staking**. Located at **pallets/staking** in the Xode repository and built against the Polkadot SDK, this pallet consolidates all staking logic into one cohesive on-chain module.

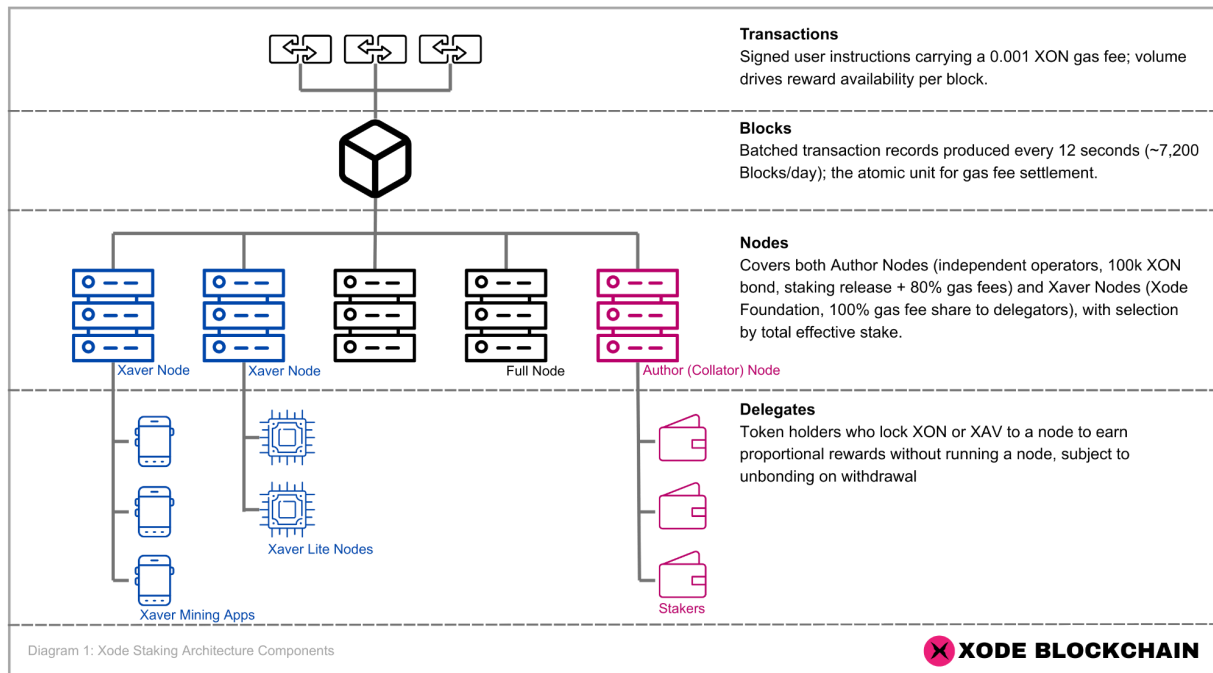
The pallet is responsible for the full lifecycle of staking on Xode, covering node registration, candidacy management, delegation handling, gas fee reward distribution, and session-based collator selection. Rather than distributing these concerns across multiple pallets, **pallet-xode-staking** acts as a unified authority for all staking states and transitions within the Xode runtime.

Key responsibilities of **pallet-xode-staking**:

- Collator candidacy and bonding - manages registration of node candidates, enforces minimum bond requirements, and maintains the on-chain candidate registry
- Xaver Node registration and automatic staking - handles Xaver Node onboarding and automatically reserves the required bond from the operator account without manual intervention
- Delegation management - tracks each delegator's locked XON token balance per node, enforces delegation constraints, and manages un-bonding queues
- Gas fee reward computation and distribution - computes per-delegator reward shares from collected gas fees and executes on-chain transfers at the end of each era

- Session-based collator selection - interfaces with the session pallet to rotate the active collator set each session, selecting nodes by total effective stake
- Token locking and reservation - coordinates with pallet-balances to lock staked and delegated XON tokens, preventing double-spend while maintaining delegator account balances

2.3 Staking Pallet Roles



Role	Function	Staking Required	Delegation
Author (Collator) Node	Block production, PoV submission, Permissionless	Yes, Minimum 10,000 XON for the node operators while at least 1 XON for the delegators	Yes, configurable commission
Xaver Node	Block production, PoV submission, Xode Foundation Authorization	No, needs Xode Foundation Authorization. Delegators staked XAV tokens.	Yes, 100% commission given through: 1. Xaver Mining App 2. Xaver Lite Node
Full Node	RPC, Archive or Bootnodes. Fully synchronized permissionless nodes for dApps and other purposes	No	No

Delegator	Permissionless token delegators for Author and Xaver nodes. 1. Locked XON for Author Nodes 2. Locked XAV for Xaver Nodes	Via delegation	No
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3. Staking Tokenomics

3.1 Token Supply

The XON staking tokens has the following supply characteristics:

Parameter	Value
Token Supply allocated for staking	122,493,748.83 XON
Author (Collator) Node Operator Reward	4%*
Author (Collator) Node Operator Locked XON Bond Provision	100,000.00 XON
Annual Reward	4,000 XON
Allocated Bond Provision	60%
Allocated Reward	40%

*The annual 4% release of XON is specifically intended to reward standard node operators who actively participate in block production and network security. This release does not apply to Xaver Nodes. Xaver Nodes operate under a separate reward mechanism and are exclusively maintained and operated by the Xode Foundation.

This separation ensures a clear distinction between the two node types and their respective incentive structures:

- **Author (Collator) Node operators** - each node operator is allocated 100,000 locked XON per node as their staking principal. On top of this, they earn rewards from two sources: (1) the 4% annual staking release applied to their 100,000 locked XON (4,000 XON per node per year), and (2) 80% of the gas fees mined from the blocks they author.
- **Xaver Nodes** - maintained and operated exclusively by the Xode Foundation. Xaver Nodes do not receive the 100,000 locked XON allocation nor any share of the annual staking release. Their reward is derived entirely from 80% of the gas fees mined, distributed proportionally to delegators based on their locked XAV token balance.

3.2 Node Capacity and Reward Depletion Model

Total XON Supply	S	
Bond allocation	S_b	$S \times 60\%$
Reward allocation	S_r	$S \times 40\%$
Bond	B	100,000 XON
Maximum number of nodes	$N_{\max}$	$\text{floor}(S_b/B)$
Total Annual Reward Outflow at Full Capacity	R	$N_{\max} \times 4\%$
Years Until Reward Pool Depletion	D	S_r / R

The depletion timeline of approximately **16.69 years** assumes all **734 node slots** are filled and active for the entire duration. In practice, governance may adjust the reward rate, bond provision, or allocation percentages, extending or modifying the schedule accordingly. Gas fee rewards earned by node operators are additional to and independent of this staking release pool.

4. Staking Model

4.1 Bonding

To participate as an Author (Collator) Node, a node must bond a minimum amount of XON. Let $B_{\min}$ denote the minimum bond. For standard collators, this is set by governance. For Xaver Nodes, the bonding occurs automatically upon node registration.

$$B \geq B_{\min}$$

4.2 Total Stake of a Node

The total effective stake of a node i , denoted M_i , is the sum of its own bond and all delegated stakes:

$$M_i = B_i + \sum(d_j) \quad \text{for all delegators } j \text{ delegating to node } i$$

$$B_i = \text{self-bond of node } i$$

$$d_j = \text{delegation amount of delegator } j$$

4.3 Authoring Selection

At the beginning of each session, the top N nodes by total effective stake are selected as active authors (collators). Let the ordered set of all candidate nodes by stake be $\{M_1 \geq M_2 \geq \dots \geq M_k\}$. The active set A is:

$$A = \{ N_i : \text{rank}(M_i) \leq N \}$$

Where N is the maximum number of active collators defined by the runtime configuration.

5. Xavier Node Staking Mechanics

5.1 Automatic Staking

Unlike Author (Collator) Nodes that require operators to manually submit bonding extrinsics, Xavier Nodes implement automatic staking at the pallet level. Upon registration, the pallet automatically reserves the required bond from the node operator's account, ensuring the node is always eligible for collator selection without manual intervention.

5.2 Delegation

Token holders can delegate XAV to any registered Xavier Node. Delegation locks the delegator's XAV tokens in a reserved state and increases the node's total effective stake, improving its likelihood of selection as an active author. Xode supports two delegation participation methods to accommodate different types of users:

Xaver Mining App

Users who prefer a fully digital participation model can obtain their delegation through the Xavier Mining App. This software application enables token holders to browse active Xavier Nodes, specify a delegation amount in XAV, and submit the delegation extrinsic directly from the app interface. Once delegated, the app provides a real-time dashboard for tracking staking rewards, node performance, commission rates, and total delegated stake. The Xavier Mining App is designed for accessibility, making staking participation available to any users regardless of technical background.

Xaver Lite Node

Users who prefer a dedicated physical device can purchase a Xavier Lite Node - a compact hardware unit purpose-built for Xode network participation. The device runs a lightweight node client that automatically connects to the Xode network, manages the delegation process on behalf of the owner, and operates independently of any general-purpose computer or mobile application. The Xavier Lite Node is designed for always-on operation, ensuring continuous uptime and reliable reward accrual. It is particularly suited to users who want a tangible, set-and-forget solution for long-term staking participation without managing software environments or private keys on shared devices.

Xaver Delegation Cost

Delegation to a Xaver Node requires the purchase of XAV tokens, which represent a user's stake share in the node. Each Xaver Node accommodates a maximum of 100,000 XAV in total delegation. XAV is available in four tiers, each priced in USD, with the top tier reserved exclusively for the Xaver Lite Node hardware product and the lower three tiers accessible through the Xaver Mining App.

Price	XAV Received	Share of Node (100,000 XAV)	% of Rewards (r)	Access Method
\$10	10 XAV	10/100,000	0.01%	Xaver Mining App
\$100	100 XAV	100/100,000	0.10%	Xaver Mining App
\$1,000	1,000 XAV	1,000/100,000	1%	Xaver Mining App
\$5,000	5,000 XAV	5,000/100,000	5%	Xaver Lite Node

A delegator's reward share is determined by their XAV balance as a proportion of the node's total XAV capacity of 100,000. The general formula for reward share percentage is:

$$\text{Reward Share (\%)} = (D_{\text{XAV}} / 100,000) \times 100$$

Multiple delegators may hold the same tier simultaneously. A node that reaches its 100,000 XAV capacity is fully subscribed and can accept no further delegations until existing delegators unstaked.

5.3 Reward Distribution

Xaver Node rewards are sourced entirely from Xode gas fee mining. The reward model differs from Author (Collator) Nodes in a critical way: 100% of the gas fees earned by an Xaver Node are passed directly to delegates. No portion is retained by the node operator. Rewards are distributed proportionally based on each delegator's locked XAV token balance equivalent percentage regardless of the number of delegates in that particular node.

Step 1: Gas Fees Collected Per Block (in XON)

Xode is a Substrate-based Polkadot parachain (Para ID 3417) that produces one block every 12 seconds. Transaction fees follow the Xode weight-based fee model, where the base fee per transaction is 0.001 XON. The total gas fees collected in a block G_{block} depend on the number of transactions T included in that block:

$$f_{\text{tx}} = 0.001 \text{ XON (base fee per transaction)}$$

$$G_{\text{block}} = f_{\text{tx}} \times T \text{ (total gas fees in XON for one block)}$$

e.g., $T = 50 \text{ txns} \rightarrow G_{\text{block}} = 0.001 \times 50 = 0.05 \text{ XON per block}$

Step 2: Convert Gas Fees from XON to USD

Gas fees are denominated in XON. To compute their USD equivalent, the current market price of XON in USD (P_{XON}) is applied. P_{XON} is a dynamic value sourced from the active exchange rate at the time of reward settlement:

$$G_{\text{usd}} = G_{\text{block}} \times P_{\text{XON}}$$

Step 3: Convert USD Value to XAV Equivalent

Since 1 XAV is pegged to 1 USD, the USD value of the gas fees collected maps directly to XAV. The total reward pool available to delegators for that block, expressed in XAV, is:

$$G_{\text{xav}} = G_{\text{block}} \times P_{\text{XON}}$$

Step 4: Individual Delegator Reward

Each delegator j receives a share of G_{xav} proportional to their locked XAV against the node's fixed capacity of 100,000 XAV. Since 1 XAV = \$1 USD and each tier represents a fixed percentage of the node, the reward formula per delegator per block is:

$$R_j = G_{\text{xav}} \times r_j$$

6. Annualized Yield Calculation

6.1 Node Operator APY

The annualized percentage yield for a node operator (i), accounting for commission and their own bond:

$$APY_i = (AR_i / B_i) \times 100\%$$

Where AR_i is the sum of commission rewards and bond-proportional rewards over one year.

6.2 Delegator APY

APY is calculated differently depending on the node type. Author (Collator) Nodes earn from the annual XON staking release and apply a commission rate. Xaver Nodes earn exclusively from gas fee mining and distribute 100% to delegators at a fixed share determined by the delegator's XAV price tier.

Author (Collator) Node

Each Author Node operator holds a locked bond of 100,000 XON and earns from the 4% annual staking release plus 80% of authored block gas fees. The node operator's base staking APY from the release allocation is:

$$r = 4\% \quad (\text{fixed annual yield on 100,000 XON bond})$$

$$\text{Annual Reward (R)} = 100,000 \times 0.04 = 4,000 \text{ XON per year}$$

Additionally, the node operator retains 80% of all gas fees from blocks they author. This component is variable and scales with network activity.

Xaver Node

Xaver Node commission is 100% of share gas fees mined flow directly to delegators. A delegator's reward is not a function of commission or total staked XON. Instead, it is determined entirely by their fixed XAV tier share of the node's 100,000 XAV capacity. The APY for a Xaver delegator is therefore gas-fee-driven and expressed as:

$$APY_j = ((f_{tx} \times T \times P_{XON} \times (D_{XAV} / 100,000) / XC_j) \times 100\%$$

Where:

- APY_j = total XAV rewards earned by delegator j in one year
- XC_j = USD cost paid for the XAV tier (\$10, \$100, \$1,000, or \$5,000)
- T = total transactions processed by the node across all blocks in a year
- $D_{XAV} / 100,000$ = delegator's fixed share of the node (0.01%, 0.1%, 1%, or 5%)

Xaver APY scales directly with network transaction volume and XON price. Higher P_{XON} and more transactions per block increase rewards proportionally. The fixed tier share ensures that all delegators at the same tier earn the same percentage of total node rewards, regardless of how many other delegators the node has.

7. Slashing & Penalties

7.1 Slashable Offences

The following behaviours result in stake slashing for a node and its delegators:

- Equivocation - producing two conflicting blocks in the same slot
- Liveness failure - missing block production obligations repeatedly beyond the defined threshold
- Invalid PoV - submitting a Proof of Validity that fails relay chain validation

7.2 Slash Calculation

For a slash of severity factor f ($0 < f \leq 1$), the total slashed amount from node i is:

$$\text{Slash}_i = S_i \times f$$

The slash is applied proportionally across the operator bond and all delegations:

$$\text{Slash}_i = B_i \times f$$

$$\text{SlashDelegator}_j = d_j \times f$$

7.3 Xavier Node Slash Protection

Xaver Nodes implement an additional layer of protection through automatic monitoring. If a Xavier Node is detected as at risk of liveness failure, the pallet can trigger an automatic reassignment to a backup node in the reserve set, minimizing the duration of liveness exposure and reducing the probability of a slashable event.

8. Unbonding & Withdrawal

8.1 Un-bonding Period

To maintain network security, staked and delegated tokens are subject to an un-bonding period U before they can be transferred. This prevents rapid stake withdrawal attacks during critical consensus windows.

$$\begin{aligned} \text{Tokens Available}(t) &= \text{Tokens Locked} && \text{if } t < t_{\text{unbond}} + U \\ \text{Tokens Available}(t) &= \text{Tokens Locked} && \text{if } t \geq t_{\text{unbond}} + U \end{aligned}$$

Where t_{unbond} is the block at which un-bonding was initiated and U is the un-bonding period in blocks.

8.2 Delegation Withdrawal

A delegator may initiate withdrawal at any time. The process:

- Step 1: Submit un-bond extrinsic - tokens enter un-bonding state, node's effective stake immediately decreases
- Step 2: Wait for un-bonding period U to elapse
- Step 3: Tokens are freed and transferable

9. Governance & Parameter Updates

9.1 Governable Parameters

The following staking parameters are adjustable through Xode's governance (Treasury Council and Technical Committee):

Parameter	Description	Default Value
S	Total XON Supply for Staking	122,493,748.83 XON
AB	Allocated Bond Provision	60%
Ar	Allocated Reward	40%
r	Annual Rate	4%
$B_{\min}$	Author Node Minimum Bond	10,000 XON

N_{max}	Maximum Number of Authors	1,000
B	Bond Provision	100,000 XON

9.2 Governance Structure

Xode's governance is managed by the Xode Foundation, a Panama-based DAO, using two collective pallets: the Treasury Council for financial decisions and the Technical Committee for protocol upgrades. The long-term roadmap includes migration to Polkadot's OpenGov model, enabling broader community participation in governance decisions including staking parameter adjustments.

10. References

- Xode Blockchain GitHub Repository: <https://github.com/Xode-DAO/xode-blockchain>
- Xode Official Website: <https://xode.net>
- Polkadot Parachain ID 3417:
<https://polkadot.js.org/apps/?rpc=wss://polkadot-rpcnode.xode.net>
- Certik Security Audit: <https://skynet.certik.com/projects/xode-blockchain>
- Substrate/Polkadot SDK: <https://github.com/paritytech/polkadot-sdk>
- CoinMarketCap Listing: <https://coinmarketcap.com/currencies/xode-blockchain/>